

Homework 2

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Problem 1. *Show that*

$$\exp \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} e^a & b \frac{e^a - e^d}{a - d} \\ 0 & e^d \end{pmatrix}$$

where the right hand side is defined by taking the limit when $a = d$. Let A be the matrix we are exponentiating.

Proof. We start by computing A^k . We claim the diagonal entries $(A_{ii})^k = (A^k)_{ii}$ and that the bottom-left corner has $A_{21}^k = 0$. The statement holds immediately for $k = 0, 1$, so assume the inductive hypothesis, and denote the off-diagonal term by A_{12}^k . Then

$$\begin{pmatrix} a^k & A_{12}^k \\ 0 & d^k \end{pmatrix} \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a^{k+1} & a^k b + A_{12}^k d \\ 0 & d^{k+1} \end{pmatrix}.$$

It suffices to compute A_{12}^k . We have the relation $A_{12}^{k+1} = a^k b + A_{12}^k d$, $A_{12}^0 = b$. We claim

$$A_{12}^k = b \sum_{i=0}^{k-1} a^{k-1-i} d^i.$$

For $k = 0$ the statement is trivial since the empty sum is 0, so assume the inductive hypothesis. We have

$$A_{12}^{k+1} = a^k b + \left(\sum_{i=0}^{k-1} a^{k-1-i} d^i \right) d.$$

$$\begin{aligned} A_{12}^{k+1} &= a^k b + \left(\sum_{i=0}^{k-1} a^{k-1-i} d^i \right) d \\ &= b \left(a^k d^0 + \sum_{i=0}^{k-1} a^{k-1-i} d^{i+1} \right) \\ &= b \left(a^{(k+1)-1-0} d^0 + \sum_{i=1}^k a^{k-1-(i-1)} d^i \right) \\ &= b \left(a^{(k+1)-1-0} d^0 + \sum_{i=1}^k a^{(k+1)-1-i} d^i \right) \\ &= b \sum_{i=0}^k a^{(k+1)-1-i} d^i. \end{aligned}$$

The statement follows by induction. We rewrite this as

$$A_{12}^k = b a^{k-1} \sum_{i=0}^k (a^{-1} d)^i.$$

Since we assume $a, d \neq 0$, this is a (finite) geometric series, and when $a \neq d$ admits the solution

$$\begin{aligned}
A_{12}^k &= ba^{k-1} \left(\frac{1 - (a^{-1}d)^k}{1 - a^{-1}d} \right) \\
&= b \left(\frac{a^{k-1} - a^{k-1-k}d^k}{1 - a^{-1}d} \right) \\
&= b \left(\frac{a^{k-1} - a^{-1}d^k}{\frac{a}{a} - \frac{d}{a}} \right) \\
&= ab \left(\frac{a^{k-1} - a^{-1}d^k}{a - d} \right) \\
&= b \frac{a^k - d^k}{a - d}.
\end{aligned}$$

We thus have

$$A^k = \begin{pmatrix} a^k & b \frac{a^k - d^k}{a - d} \\ 0 & d^k \end{pmatrix}.$$

We now compute the matrix exponential

$$e^A = \sum_{n=0}^{\infty} \frac{1}{n!} A^n = \begin{pmatrix} \sum_{n=0}^{\infty} \frac{a^n}{n!} & \sum_{n=0}^{\infty} \frac{b}{n!} \frac{a^n - d^n}{a - d} \\ 0 & \sum_{n=0}^{\infty} \frac{d^n}{n!} \end{pmatrix} = \begin{pmatrix} e^a & \sum_{n=0}^{\infty} \frac{b}{n!} \frac{a^n - d^n}{a - d} \\ 0 & e^d \end{pmatrix}.$$

We note that

$$\sum_{n=0}^{\infty} \frac{b}{n!} \frac{a^n - d^n}{a - d} = \frac{b}{a - d} \sum_{n=0}^{\infty} \frac{a^n - d^n}{n!} = \frac{b}{a - d} \left(\sum_{n=0}^{\infty} \frac{a^n}{n!} - \sum_{n=0}^{\infty} \frac{d^n}{n!} \right) = \frac{b}{a - d} (e^a - e^d).$$

Here we used the fact that $n! \rightarrow \infty$ faster than $a^n - d^n$, hence the sum can be shown to converge absolutely, and thus we may commute the terms in the infinite sum to distribute over the subtraction. Thus

$$e^A = \begin{pmatrix} e^a & b \frac{e^a - e^d}{a - d} \\ 0 & e^d \end{pmatrix}.$$

To determine e^A when $a = d$, we take the limit $a \rightarrow d$ and apply L'Hopital's rule by differentiating with respect to a to get

$$\lim_{a \rightarrow d} \frac{e^a - e^d}{a - d} = \lim_{a \rightarrow d} \frac{e^a}{1} = e^a.$$

Alternatively, we could have used

$$A_{12}^k = ba^{k-1} \sum_{i=0}^k (a^{-1}a)^i = ba^{k-1} \sum_{i=0}^k 1 \equiv kba^{k-1}.$$

Then the top right entry of the matrix exponential would have been

$$\sum_{n=0}^{\infty} \frac{nba^{n-1}}{n!} = \frac{d}{da} \sum_{n=0}^{\infty} \frac{ba^n}{n!} = \frac{d}{da} be^a = be^a.$$

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Problem 2. Consider

$$X = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Compute e^{tX} and e^{tY} by diagonalization. Visualize the curves $e^{tX} \cdot v$ and $e^{tY} \cdot v$ for $v \neq 0$.

Proof. To diagonalize a matrix, we compute its eigenvalues and eigenvectors. We start by diagonalizing X .

$$\det \begin{pmatrix} \lambda - 1 & -2 \\ 2 & \lambda - 1 \end{pmatrix} = \lambda^2 - 2\lambda + 5 = 0$$

$$\implies \lambda = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i.$$

$$\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (1+2i)x \\ (1+2i)y \end{pmatrix} \implies \begin{pmatrix} x+2y \\ -2x+y \end{pmatrix} = \begin{pmatrix} x+2ix \\ y+2iy \end{pmatrix}.$$

We obtain from $x+2y = x+2ix$ that $2y = 2ix$, hence $y = ix$ and $x = -iy$. Set $x = 1$ to obtain the first eigenvector $v_1 = (1, i)$.

$$\begin{pmatrix} x+2y \\ -2x+y \end{pmatrix} = \begin{pmatrix} x-2ix \\ y-2iy \end{pmatrix}.$$

We obtain $2y = -2ix$, hence $y = -ix$. Set $x = 1$ to obtain the second eigenvector $v_2 = (1, -i)$. The inverse of $\begin{pmatrix} v_1 & v_2 \end{pmatrix}$ is given by

$$\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}^{-1} = \frac{1}{-2i} \begin{pmatrix} -i & -1 \\ -i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$

We then have

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} 1+2i & 0 \\ 0 & 1-2i \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+2i & 1-2i \\ -2+i & -2-i \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 4 \\ -4 & 2 \end{pmatrix} = X.$$

Thus the matrix is properly diagonalized. Write $X = ADA^{-1}$ with A, D defined as the relevant above matrices. We then compute the matrix exponential

$$e^{tX} = Ae^{tD}A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} e^{t(1+2i)} & 0 \\ 0 & e^{t(1-2i)} \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} =$$

$$\frac{1}{2} \begin{pmatrix} e^{t(1+2i)} + e^{t(1-2i)} & -i(e^{t(1+2i)} - e^{t(1-2i)}) \\ i(e^{t(1+2i)} - e^{t(1-2i)}) & e^{t(1+2i)} + e^{t(1-2i)} \end{pmatrix}.$$

Euler's formula provides some convenient simplifications.

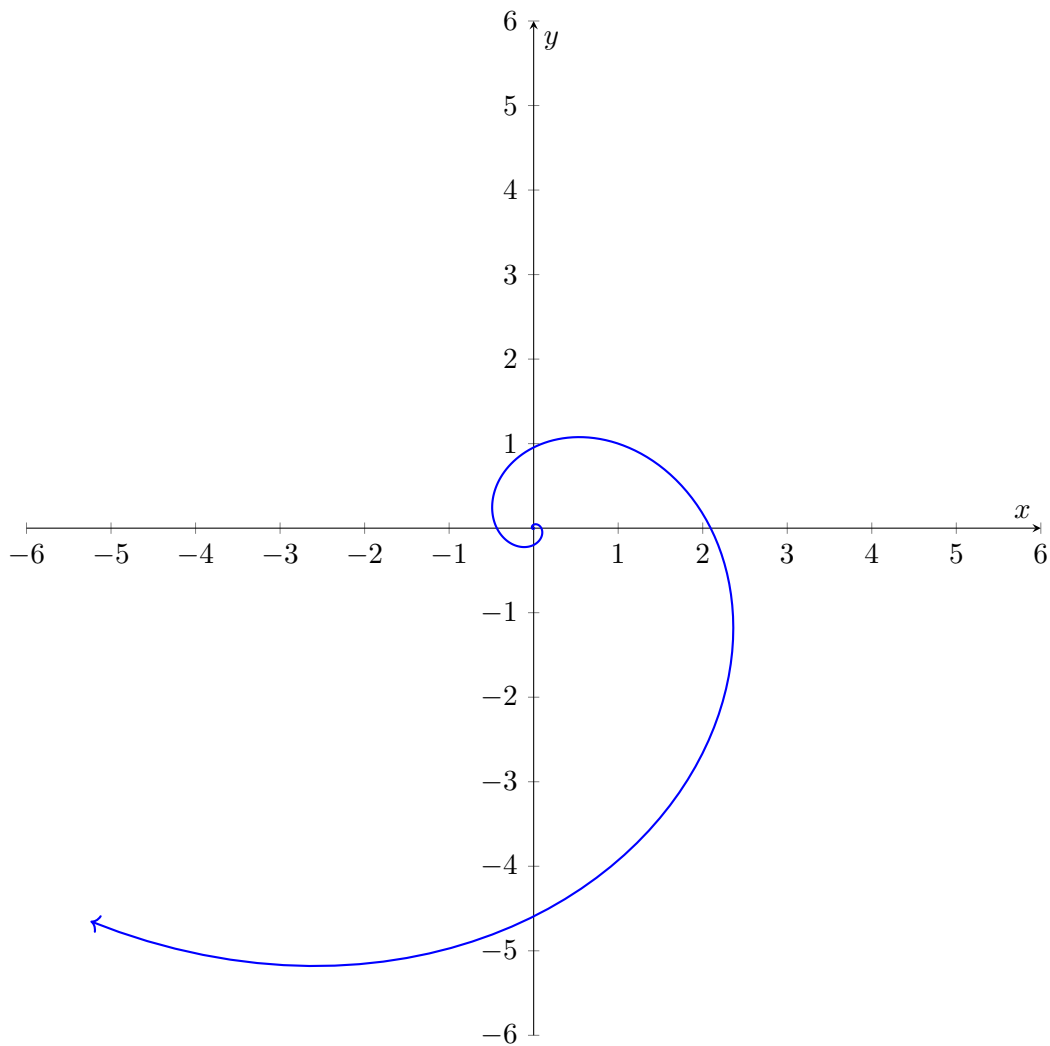
$$e^{t(1+2i)} + e^{t(1-2i)} = 2e^t \cos(2t), \quad e^{t(1+2i)} - e^{t(1-2i)} = 2ie^t \sin(2t).$$

Then

$$e^{tX} = \begin{pmatrix} e^t \cos(2t) & e^t \sin(2t) \\ e^t \sin(2t) & e^t \cos(2t) \end{pmatrix}.$$

We fix $v = (1, 1)$ and plot

$$e^{tX} \cdot v = \begin{pmatrix} e^t (\cos(2t) + \sin(2t)) \\ e^t (\cos(2t) - \sin(2t)) \end{pmatrix}.$$



We now repeat the process for Y , showing much less work. We obtain

$$Y = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}.$$

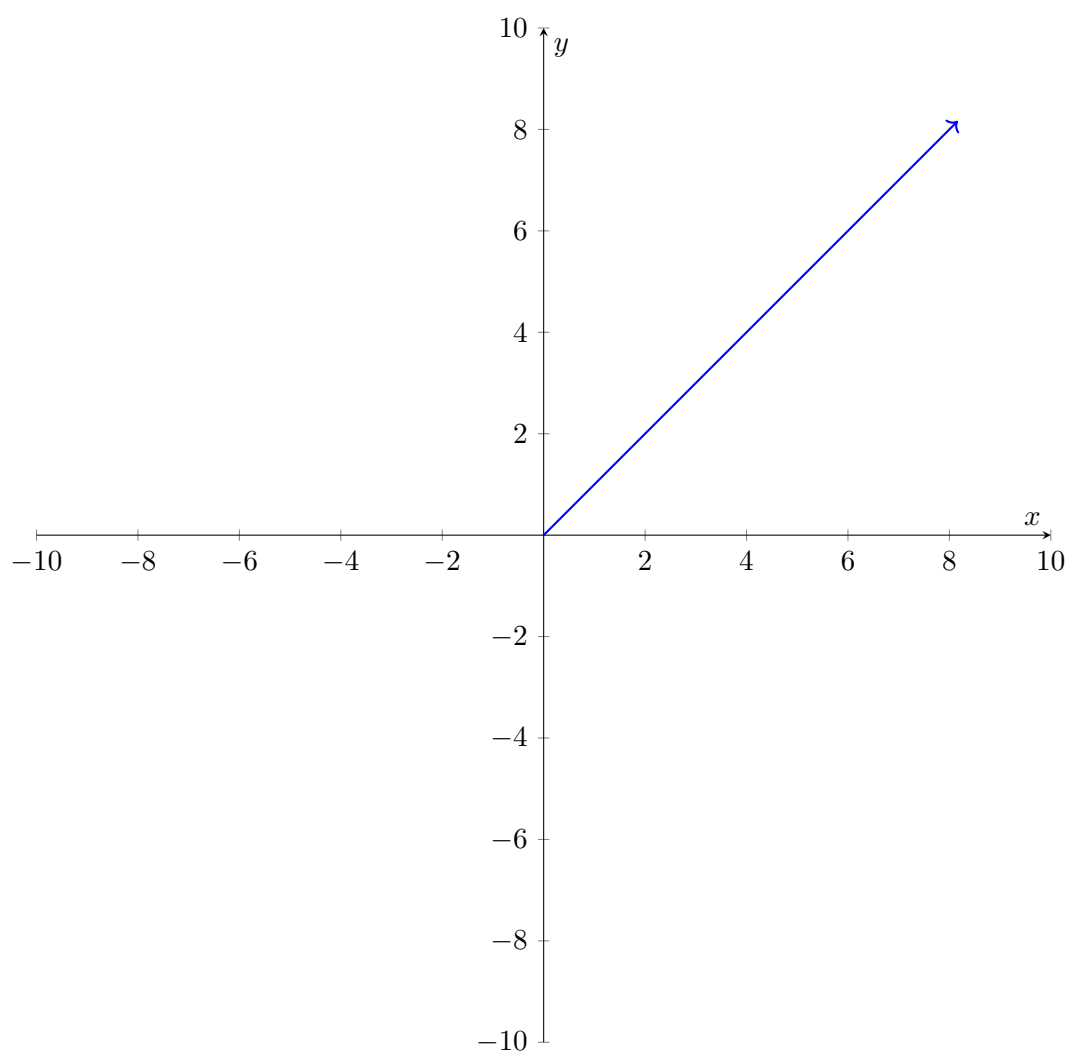
Then

$$e^{tY} = \frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{3t} \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^{-t} & e^{3t} - e^{-t} \\ e^{3t} - e^{-t} & e^{3t} + e^{-t} \end{pmatrix}.$$

Again choosing $v = (1, 1)$, we obtain

$$e^{tY} \cdot v = \begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix}.$$

Plotting:



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